

Feral Rabbits

Oryctolagus cuniculus (aka European rabbit)



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Overview:

Feral rabbits are native to Europe and northwestern Africa but have been introduced to every continent except Antarctica.¹ They were first introduced to North America in the 18th century by European settlers bringing them as food and fur. Now they can be found in the wild from introduction by pet owners who can no longer care for them. Their presence can be very harmful to environments. Many native species in the Leporidae family can be mistaken for feral rabbits, like mountain cottontails, snowshoe hares, and white-tailed jackrabbits.

Rabbits can spread and reproduce rapidly as females, known as does, can have up to 3 litters each year and begin breeding as young as 3 months old.^{1,2} A single litter can consist of 4-12 kits (baby rabbits), with an average litter size of 5. Rapid reproduction is a significant factor in the population explosion of feral rabbits. Feral rabbits can live up to 10 years in captivity, but many only live about 12-15 months in the

wild. Due to outdoor conditions, released rabbit populations deal with diseases, predators, malnutrition, and extreme weather. While many survive long enough to reproduce, they live a much shorter and more dangerous life in the wild than if they had been kept as pets.

Distribution:

Feral rabbits have been reported across Alberta in both urban and rural environments.

Habitat:

Thrive in urban areas, parks, farmlands, and grasslands. They prefer areas with thick cover to hide in and soft ground to create shallow burrows for shelter but are highly adaptable.

Identification:

Feral rabbits come in a variety of colours from white, black, brown to a combination and do not change colour with the seasons. Typically have a beige ring around the eyes. Ears are smaller and legs are shorter than hares. Usually weigh between 2-5 lbs and measure 35-50 cm in length.

Economic Impact:

Overgrazing can become an issue in areas where feral rabbits are present. This can impact agriculture from crop revenue losses, as well as urban gardens and landscaping maintenance costs from rabbits gnawing on trees and shrubs and stripping bark. Due to their high rate of reproduction, once they have been established it is very hard to control rabbit populations, and control methods require large amounts of time and money to achieve. The Town of Canmore spent up to \$50,000 per year hiring contractors to trap and euthanize feral rabbits.⁵

Environmental Impact:

Feral rabbits can have a detrimental impact on native flora by overgrazing, leading to soil erosion and biodiversity loss. They gnaw on the bark of woody plants in the winter when no green growth is available and can do significant damage by girdling young stems or nipping off all the buds and shoots. Their burrowing can cause land degradation and erosion. These adaptable herbivores have become



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invasive in the local ecosystem, outcompeting native herbivores for food resources and altering vegetation dynamics. Their prolific breeding exacerbates the problem, leading to overgrazing of plant species. This disruption can result in the loss of biodiversity, affecting not only plants but also the insects and animals dependent on them. Their large numbers often support increased populations of predators that can then harm native prey species, and they experience higher than normal predation.

Sociological Impacts:

Feral rabbits can carry diseases such as tularemia and myxomatosis, which may pose a risk to other wildlife and domestic pets. Alberta populations have been dealing with outbreaks of rabbit haemorrhagic disease³, a highly lethal and infectious virus that could begin to spread to native rabbit and hare species. If you see a potentially diseased rabbit, please report it to your local veterinarian or the Government of Alberta Wildlife Disease Unit.

Prevention:

Don't Let It Loose! The best way to prevent further spread of feral rabbits is to spread awareness about the dangers of releasing pets into the wild. Releasing a pet of any kind into the wild is illegal in Alberta and can be subject to a fine. Report feral rabbit sightings and issues to your municipality or on EDDMapS.

Areas can also be modified to discourage and exclude rabbits.⁴ By removing unnecessary undergrowth, a key area for shelter, it will discourage rabbits from moving in. Mechanical barriers are most effective at protecting vegetation. Mesh fences should be at least 1m high for vegetation, and 1m above the potential snow line for protecting trees or shrubs. Fences should also be set 7-10 cm into the ground to prevent rabbits from digging underneath. Hardware cloth can make a good barrier as well and should follow the same guidelines. Avoid placing any barriers directly against vegetation as rabbits can reach through the holes to feed.

Control:

Native rabbits and hares can be hunted at any time of year in appropriate hunting areas of Alberta without a license.⁴ It is illegal to poison rabbits or use leg-hold traps to catch rabbits in Alberta. Live trapping can be used to remove rabbits from an area but must be done humanely. Wooden box traps work well and should be placed in areas sheltered from the weather and should be checked regularly (at least once per 24 hours if food and water are provided in the trap). Repellents can be used to make plants distasteful to rabbits with varied success depending on how much competition there is for food in your area. Animals and humans cannot consume repellent-treated plants. Two chemical repellents are registered for use in Alberta, both sold under the name SKOOT.⁴



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REFERENCES:

1. Tislerics, A. *Oryctolagus cuniculus* European rabbit. Animal Diversity Web. University of Michigan Museum of Zoology. https://animaldiversity.org/accounts/Oryctolagus_cuniculus/. Accessed: November 2023
2. *Oryctolagus cuniculus*. Global Invasive Species Database. <http://www.iucngisd.org/gisd/speciesname/Oryctolagus+cuniculus>. Accessed: November 2023.
3. Alberta Environment and Parks. 2022. Rabbit Haemorrhagic Disease. Accessed November 2023 from: <https://open.alberta.ca/dataset/c990a9c2-cc7a-4471-aa4b-ce73dd1f8f7f/resource/3af774d2-ae6c-4256-8953-b0717dac20c0/download/aep-rabbit-haemorrhagic-disease-2022-fall.pdf>
4. Alberta Agriculture Food and Rural Development. Revised 2005. Control of Rabbits and Hares. Accessed online November 2023 from: <https://open.alberta.ca/dataset/4f68df26-b8d7-4c57-91c5-092f4a28bd90/resource/a8b92fbc-78b4-46db-882b-3aca1aa98a37/download/2005-684-12.pdf>
5. CBC News Calgary. 2023. Accessed online April 25, 2024 from: <https://www.cbc.ca/news/canada/calgary/canmore-feral-rabbits-rhd-rabbit-hemorrhagic-disease-1.7054595>